

Project Planning Backlog: ShopEZ Stock Trading Platform

This document outlines the Product Backlog, Sprint Schedule, Estimation, and Velocity metrics for the MERN-based ShopEZ Stock Trading Platform.

Project Parameters

- **Project Name:** ShopEZ Stock Trading Platform
- **Date:** 17 June 2026
- **Team ID:** MERN-Group-04
- **Sprints:** 4 Sprints (6 working days per sprint)
- **Starting Velocity:** 9.0 Story Points per sprint (Daily Output Rate of 1.5 SP/day)

Product Backlog, Sprint Schedule, and Estimation

Use the table below to track user stories, priorities, estimations, and sprint assignments:

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Assigned Developer
Sprint-1	User Authentication & Profiles	USN-1	As a user, I can register for an account by entering my name, email, password, and confirming password, automatically receiving a starting virtual balance of \$10,000.	3	High	Dev-01
Sprint-1	User Authentication & Profiles	USN-2	As a registered user, I can log in securely using my email and password to receive a JWT session token and access the trading dashboard.	3	High	Dev-02
Sprint-1	User Authentication & Profiles	USN-3	As an authenticated user, I can view my current account profile and live virtual balance in the navbar.	2	Medium	Dev-02
Sprint-2	Market Dashboard & Stock Browsing	USN-4	As a user, I can view a live dashboard showing stock listings, market updates, and featured tickers with Outfit/Inter typography and soft pastel UI.	3	High	Dev-01
Sprint-2	Market Dashboard & Stock Browsing	USN-5	As a user, I can search for active listings by ticker symbol or company name to quickly locate specific stocks.	2	Medium	Dev-02

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Assigned Developer
Sprint-2	Market Dashboard & Stock Browsing	USN-6	As a developer, I can implement a background price simulator that fluctuates listed stock prices by -1.2% to +1.2% every 5 seconds.	5	High	Dev-01
Sprint-3	Simulated Trading Flow	USN-7	As a trader, I can view detailed stock charts (30-day historical performance) and execute buy orders that validate my virtual cash balance before deducting funds.	5	High	Dev-01
Sprint-3	Simulated Trading Flow	USN-8	As a trader, I can execute sell orders that validate my owned stock quantity before crediting my balance.	4	High	Dev-02
Sprint-3	Simulated Trading Flow	USN-9	As a trader, I can view my transactional history logs showing timestamps, buy/sell flags, execution prices, and share volumes.	2	Medium	Dev-02
Sprint-4	Portfolio Tracking	USN-10	As a trader, I can view my holdings aggregated on a portfolio screen displaying cost basis, current market value, and overall profit/loss with soft-red or mint-green indicators.	3	High	Dev-01
Sprint-4	Administration & Moderation	USN-11	As an administrator, I can view a dashboard showing all registered users, manage active stock tickers (create, edit, delete), and inspect the global audit trail.	4	High	Dev-02
Sprint-4	Dynamic Updates & Alerts	USN-12	As a user, I can see real-time updates and success/error status alerts in the viewport when trades occur or prices fluctuate.	2	Low	Dev-01

Project Tracker, Velocity & Burndown Chart

Velocity Calculations

- **Total Story Points (SP):** 38 Story Points completed.
- **Sprint Duration:** 6 working days per sprint.
- **Total Project Duration:** 24 working days (4 Sprints).
- **Average Velocity:** 9.5 Story Points per sprint iteration.
- **Daily Output Rate:** 1.58 Story Points per day.

Daily Output Rate = $\frac{\text{Velocity}}{\text{Sprint Duration}} = \frac{9.5 \text{ SP}}{6 \text{ Days}} \approx 1.58 \text{ SP/Day}$

Burndown Chart Metrics

The burndown chart tracks remaining story points across the 24-day project timeline:

- **Day 0:** 38 SP remaining
- **Day 6 (End of Sprint 1):** 30 SP remaining (USN-1, USN-2, USN-3 completed)
- **Day 12 (End of Sprint 2):** 20 SP remaining (USN-4, USN-5, USN-6 completed)
- **Day 18 (End of Sprint 3):** 9 SP remaining (USN-7, USN-8, USN-9 completed)
- **Day 24 (End of Sprint 4):** 0 SP remaining (USN-10, USN-11, USN-12 completed)